Design, Implementation and Cloud Evaluation of a Fault-Tolerant, Horizontally Sharded Durable Execution Engine

A PROJECT REPORT

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ABSTRACT

A long-running workflow, such as an order, a payment or a multi-step software agent, changes the state of several external services, and a crash part way through it either loses paid-for work or repeats work that must not happen twice. Durable execution engines prevent this by journalling every workflow decision and recovering by deterministic replay, but production engines depend on four to six infrastructure services. This report presents *durexec*, a durable execution engine in Go whose only dependency is PostgreSQL. Workflows are hashed to a fixed set of shards; each shard is owned through a lease held in a database row, and a fencing token re-checked inside every write transaction stops a node that has lost its lease from committing. A journalled semantic check lets the engine recognise a reworded retry of an action already taken by a language-model agent. On Amazon EKS with three engine nodes and one PostgreSQL host, a closed-loop workload reached 131 workflows per second at a 99th-percentile latency of 3.1 s, limited by database CPU, and the sustainable open-loop rate was about 100 workflows per second. Workers, shards and engine replicas changed throughput by at most 19%, and extra replicas gave no gain while the database was the bottleneck. In sixteen fault-injection trials, covering engine termination, a 35 s process freeze, loss of database connections, a node drain and a database reboot, every accepted workflow completed, no journal invariant was violated and every write by a stale owner was rejected. On a corpus of 213 request pairs, similarity of sentence embeddings alone suppressed most similar but distinct requests, whereas comparing the deciding fields exactly separated every pair.

Keywords: durable execution, event sourcing, deterministic replay, leases, fencing tokens, sharding, fault tolerance, PostgreSQL.

TABLE OF CONTENTS

Page No.

ACKNOWLEDGEMENTS i

ABSTRACT ii

LIST OF FIGURES vi

LIST OF TABLES vii

CHAPTER 1 – INTRODUCTION 1

1.1 Background 1

1.2 Motivation 3

1.3 Problem Statement 4

1.4 Objectives 4

1.5 Scope and Limitations. 4

1.6 Contributions 5

1.7 Organisation of the Report 6

CHAPTER 2 – LITERATURE SURVEY 7

2.1 Durable Execution and Stateful Serverless Workflows 7

2.2 Partitioning Work Across Machines 8

2.3 Leases, Coordination and Fencing 9

2.4 Long-Running Transactions and Sagas 10

2.5 Failure Modes of Agent Systems 10

2.6 Semantic Similarity and Caching 11

2.7 Comparison of Related Systems 12

2.8 Research Gap 12

CHAPTER 3 – REQUIREMENTS AND ANALYSIS 13

3.1 Functional Requirements 13

3.2 Non-Functional Requirements 13
3.3 Software and Hardware Requirements 13
3.4 Assumptions and Constraints 13
3.5 Analysis of the Chosen Approach 16
CHAPTER 4 – SYSTEM DESIGN 18
4.1 High-Level Design 18
4.2 Low-Level Design 22
CHAPTER 5 – IMPLEMENTATION 28
5.1 Code Organisation 28
5.2 The Fenced Commit 29
5.3 Task Queue 29
5.4 Lease Manager 30
5.5 Workflow Execution 30
5.6 Semantic and Hybrid Deduplication 30
5.7 API, Routing and Observability 31
5.8 Demonstration Applications 32
5.9 Deployment 32
CHAPTER 6 – TESTING AND EVALUATION METHOD 33
6.1 Unit and Integration Tests 33
6.2 Fault Injection at Crash Points 34
6.3 Local Cluster Tests 34
6.4 Cloud Test Bed 35
6.5 Measurement Procedure 35
6.6 Fault-Injection Procedure 37
CHAPTER 7 – RESULTS AND DISCUSSION 39
7.1 Latency Floor 39
7.2 Idle Overhead 40

7.3 Closed-Loop Capacity. 40
7.4 Open-Loop Capacity and Overload 42
7.5 Sensitivity to Workers, Shards and Replicas 44
7.6 Recovery from Faults 45
7.7 Correctness Summary 47
7.8 Semantic Deduplication 47
7.9 Threats to Validity 48
CHAPTER 8 – CONCLUSION AND FUTURE WORK. 49
8.1 Conclusion 49
8.2 Future Work 50
REFERENCES 51

LIST OF FIGURES

4.1 System architecture 19

4.2 Two-level mapping of workflows to shards and nodes 21

4.3 Lease expiry and fencing of a stale owner 22

4.4 The workflow task loop 23

6.1 Cloud test bed on AWS 36

6.2 Timing of a fault trial 38

7.1 Components of the latency floor 39

7.2 Idle database load. 40

7.3 Closed-loop throughput and latency. 41

7.4 Resource use against concurrency 42

7.5 Database wait events. 42

7.6 Open-loop offered against completed load. 43

7.7 Sensitivity to workers, shards and replicas. 44

7.8 Recovery times by fault 46

LIST OF TABLES

1.1 Terminology used in this report. 2

1.2 Project objectives 5

2.1 Comparison of related systems 12

3.1 Functional requirements 14

3.2 Non-functional requirements 15

3.3 Hardware used for development and evaluation 15

4.1 Persistent data model 22

4.2 Crash windows for one activity execution 25

5.1 Main packages of the engine 28

5.2 HTTP API 31

6.1 Distribution of test functions by package. 34

6.2 Evaluation stages 36

7.1 Closed-loop results (r3/S16/w4x8). 41

7.2 Open-loop results (three trials per rate). 43

7.3 Capacity at $c = 256$ under parameter changes (mean of three trials) . . 45

7.4 Fault recovery under 65.6 wf/s open-loop load 45

7.5 Deduplication on the 213-pair corpus at the shipped thresholds 47

CHAPTER - 1

INTRODUCTION

1.1 Background

Much of the software that runs a business is not a single request answered in milliseconds but a process that unfolds over seconds, hours or days. Placing a food order charges a card, reserves stock, notifies a restaurant and assigns a driver. Settling an invoice debits one ledger, credits another and files a tax record. An agent built on a large language model that is asked to arrange travel plans an itinerary, calls a booking service, waits for a person to approve the result and then confirms it. Each of these is a *workflow*: a sequence of steps, spread over several independent services, in which every step depends on the results of the steps before it and most steps change the state of some system outside the workflow itself.

Two properties make workflows hard to run reliably. A workflow is stateful, because step four only makes sense in the light of what steps one to three decided. Its steps also have external side effects that the workflow cannot withdraw: once a payment provider has moved money, no local rollback can move it back. A database transaction has neither difficulty, since it is short enough to hold its locks throughout and its effects are confined to a system that can undo them. A workflow has both.

As a result, an ordinary crash becomes a correctness problem. Suppose the order workflow is interrupted after the card has been charged and the stock reserved, but before the restaurant has been told. A system that has kept no durable record of its progress has two choices when it restarts. It can run the workflow again from the beginning, charging the card a second time, or it can abandon the workflow, keeping the customer's money for an order that will never be delivered. Without a record of what was already done, there is no correct third option.

The usual responses in practice are retry loops, periodic jobs that poll a status column, and message queues whose consumers record progress in application tables. Each works for the workflow it was written for, and each must be written again for the next one.

Durable execution is the name given to systems that solve the problem once for every workflow built on them [1, 2]. The idea on which they rest, and on which this report rests, is that *every decision a workflow makes is appended to a log before it is acted on, and recovery is performed by running the workflow's own code again against that log*. Completed steps return their recorded results instead of executing a second time, so the re-executed code arrives at the point the failed process had reached. Because nothing private to one machine is part of the workflow's state, the recovery can take place on a different machine.

Table 1.1 fixes the terms used throughout the report.

Table 1.1: Terminology used in this report

Term	Meaning
Workflow	Orchestration code that decides what happens next and performs no input or output itself. It must be deterministic, because it is re-executed on every recovery.
Activity	One step with an external side effect, such as charging a card or calling an API. It may be non-deterministic and runs at least once.
Event	An immutable record appended to the log describing something that has already happened.
History	The ordered events of one workflow run; the complete state of that run.
Replay	Re-executing a workflow against its history so that completed steps resolve from the log.
Shard	One of a fixed number of partitions. A workflow belongs to one shard for its whole life.
Lease	A time-limited claim on a shard, held in a database row and renewed periodically, that makes one node the owner of the shard.
Fencing token	A counter that increases whenever ownership of a shard changes; a write is accepted only with the current value.
Saga	A multi-step transaction that, if it fails part way, is undone by running a compensating action for each step that had succeeded.

Industry has adopted this approach widely. Uber's Cadence and its successor Temporal run very large numbers of workflows for ride and delivery systems [1]; Microsoft's Durable Functions provide the same model for serverless code [3]; and the DBOS project argues that durable execution belongs inside the database rather than in a separate service [4, 5]. The internal mechanisms of such an engine, namely how the log is partitioned, how state is reconstructed deterministically and how ownership of a par-

tition moves safely between machines when one fails, are less often built from first principles. This project builds one, and evaluates it on a cloud cluster under load and under injected failures.

1.2 Motivation

1.2.1 Existing engines depend on several services

The production engines surveyed for this work each require four to six cooperating infrastructure components. Netflix's Conductor uses a database, a search cluster, a queue, a server tier and a worker fleet. Temporal runs four internal services over a database and, for visibility queries, a search cluster [1]. Meta's Shard Manager, which assigns partitions across very large fleets, relies on ZooKeeper for coordination [6, 7]. Each additional component is a separate failure domain that must be configured, monitored and understood during an incident, and for a small deployment the orchestration system can cost more to operate than the application it serves. This project asks whether the same guarantees can be provided with a single dependency.

1.2.2 Partitioning and simplicity are usually traded off

The simple systems are not partitioned and the partitioned systems are not simple. DBOS obtains durable execution with very few moving parts but runs on one node [4]; Temporal and Shard Manager partition work across machines but add a coordination service or a multi-service control plane to decide which machine owns what. Our position is that the two need not be in tension, because a relational database already provides the linearizable compare-and-swap that partition ownership requires [8, 9]. What is needed is a careful use of one row per shard rather than a consensus cluster [10].

1.2.3 Agents defeat byte-level deduplication

Durable execution engines prevent duplicate side effects by recognising a repeated request, and they recognise it by comparing bytes: a hash, an idempotency key or an exact match on recorded arguments. This is sound when the arguments are fixed at the moment a step is first scheduled. It is not sound when the caller is a language-model agent, which after a crash *re-plans* rather than replays: an instruction recorded as "BLR→DEL Friday morning" may be issued again as "Bangalore to Delhi, Friday

AM”. The two strings hash differently and the flight is booked twice. Step repetition of this kind appears as a recurring category in empirical studies of multi-agent failures [11] and in Microsoft’s taxonomy of agentic failure modes [12].

1.3 Problem Statement

The problem addressed in this work is to design and implement a horizontally sharded durable execution engine that runs long, multi-step workflows to completion despite process crashes, node failures and partial failures, using event sourcing, deterministic replay, idempotent activity execution and lease-based ownership; that is deployable on a small cluster with PostgreSQL as its only infrastructure dependency; that detects semantically equivalent repeated actions issued by a recovering agent before the side effect occurs a second time; and to measure its throughput, its sensitivity to its main configuration parameters, and its recovery from injected faults on real cloud infrastructure.

Two limits are part of the statement. The engine targets correctness and fault tolerance at a moderate, measurable scale rather than the throughput of large production deployments. Ownership is obtained from a lease in a linearizable database rather than from a consensus protocol, which makes the database the availability floor of the system.

1.4 Objectives

The work has eight objectives, listed in Table 1.2 with the mechanism that realises each and the chapter in which it is evaluated.

1.5 Scope and Limitations

The scope is the engine, two demonstration applications built on it, and the test bed used to evaluate it. The main limitations are stated here so that the design and the results can be read with them in mind.

1. The engine provides at-least-once execution of activities with strong deduplication, not exactly-once execution. A crash after a side effect has happened but before it has been recorded cannot be told apart from a crash before the side effect. The engine supplies a stable deduplication key that the downstream service can use to close this window.

Table 1.2: Project objectives

No.	Objective	Mechanism	Evaluated
1	Record every state transition as an immutable event so that a run can be reconstructed exactly	Append-only event log	Ch. 6, 7
2	Resume a failed run from its history rather than from the start	Deterministic replay	Ch. 6, 7
3	Never repeat an external side effect across retries	Stable deduplication keys	Ch. 6
4	Partition execution across nodes	Fixed shards, consistent-hash ring	Ch. 7
5	Lose no progress when a shard owner fails	Leases with fencing tokens	Ch. 7
6	Undo multi-step transactions that fail part way	Saga compensation	Ch. 6
7	Detect reworded duplicate agent actions	Journalled semantic verdict	Ch. 6
8	Measure capacity, sensitivity and recovery on cloud hardware	EKS test bed, fault injector	Ch. 7

2. Ownership relies on PostgreSQL as the single source of truth. The cloud evaluation used one database host with no standby, so it says nothing about database high availability.

3. The event log is never garbage collected; retention is future work.

4. Semantic deduplication is opt-in and conservative. A false negative is no worse than exact matching, but a false positive would suppress an action the user wanted, so the semantic mode must be requested per activity and every verdict is journalled.

5. All cloud measurements come from one availability zone on burstable (“flex”) instance types, over individual runs of a few minutes. They establish the shape of the behaviour on this configuration, not absolute capacity for other hardware.

1.6 Contributions

The contributions of this work are as follows.

1. A durable execution engine in which event sourcing, replay, a durable task queue, timers, signals, sagas and shard failover are implemented over PostgreSQL alone, with no message broker or coordination service.

2. Fencing at the persistence layer: every transaction that writes shard state re-reads the shard's fencing token with `SELECT ... FOR UPDATE` and commits only if the token is still the writer's. On the cloud test bed this rejected every write attempted by a stale owner after a process freeze.
3. A two-level mapping in which workflows hash to a fixed set of shards and shards are assigned to nodes by a consistent-hash ring, so that a change of membership moves ownership but no data.
4. A semantic idempotency check that compares the meaning of an agent's intended action with the actions already in the run's history and journals the verdict, keeping replay deterministic and auditable.
5. A cloud evaluation on Amazon EKS covering idle overhead, latency floor, closed- and open-loop capacity, sensitivity to workers, shards and replicas, and recovery from six classes of injected fault, together with the evaluation harness and its validity checks.

1.7 Organisation of the Report

Chapter 2 reviews related work on durable execution, partition management, ownership and fencing, sagas, and semantic similarity, and identifies the gap this work addresses. Chapter 3 states the requirements and the analysis that led to the chosen approach. Chapter 4 presents the high-level and low-level design. Chapter 5 describes the implementation and Chapter 6 the testing methodology, including the cloud test bed and the fault-injection procedure. Chapter 7 reports and discusses the results, and Chapter 8 concludes and outlines future work.

CHAPTER - 2

LITERATURE SURVEY

This chapter reviews the work on which the engine is built. Six bodies of literature are relevant: durable execution and stateful serverless systems; the partitioning of work across machines; leases, coordination and fencing; long-running transactions and sagas; the failure modes of agent systems built on language models; and semantic similarity and caching. Each section closes with the point taken from it into the design. The chapter ends with a comparison of the closest systems and a statement of the gap.

2.1 Durable Execution and Stateful Serverless Workflows

The principle that a computation can be recovered by re-executing it against a log of its own decisions is older than the term durable execution. Lamport showed that a deterministic state machine fed the same ordered inputs reaches the same state [13], and the write-ahead log of classical transaction processing applies the same reasoning to database pages [14]. Log-centred systems such as Kafka later made an ordered, durable log the main interface between services [15]. Durable execution engines apply the idea at the level of application code: the inputs to the state machine are the results of the workflow's external calls, and the log is the workflow's event history.

Temporal, which grew out of Uber's Cadence, is the most widely used engine of this kind [1]. Workflow code is written in an ordinary programming language under a determinism contract; each external call is recorded as a pair of scheduled and completed events, and recovery replays the code against the history. Temporal partitions histories into a fixed number of shards, and a membership service assigns shards to history service hosts. The architecture is proven at large scale, but a deployment comprises frontend, history, matching and worker services over a database, with an optional search cluster for visibility queries. Restate takes a similar programming model but embeds its own log and storage in a single binary [2].

Durable Functions brought the model to serverless platforms. Burckhardt *et al.* give a formal semantics for orchestrations, entities and replay, and prove that replay is correct

when the orchestration code is deterministic [3]. Netherite replaced the original storage provider with partitioned, speculatively committed logs and reported large gains in throughput and latency [16]. Ambrosia earlier proposed *virtual resiliency*, in which logging and deterministic replay make a distributed application behave as if it never failed, at the cost of requiring every component to be deterministic [17].

Two research systems from the serverless community address the same problem with a shared log. Beldi provides exactly-once semantics and transactions for stateful serverless workflows by logging every read and write in a linked data-access log kept in a key-value store [18]. Boki builds a shared log into the function runtime and offers durable workflows, object storage and message queues as libraries over it [19]. Both achieve strong guarantees, and both depend on a purpose-built logging layer.

DBOS argues the opposite direction: that the state of an operating system, including the state of running programs, belongs in a database [4]. Its durable workflows checkpoint each step into PostgreSQL, so that recovery needs nothing other than the database. R3 extends the approach to record and replay of database-backed applications, so that a past execution can be reproduced and modified retroactively [5]. DBOS is the system closest in spirit to this project. It does not, however, partition workflows across nodes with explicit ownership; recovery is driven by the database and the executor that originally ran the workflow.

Taken into the design. The workflow and activity split, the determinism contract and the recovery-by-replay model come from Temporal and Durable Functions. The decision to use PostgreSQL as the only store follows DBOS. The engine differs from all of them in combining a single-dependency deployment with explicit, fenced shard ownership.

2.2 Partitioning Work Across Machines

A system that runs on more than one machine must decide which machine is responsible for each piece of state. Consistent hashing, introduced by Karger *et al.* for web caches [20], maps keys and machines onto the same circle so that adding or removing a machine moves only the keys adjacent to it. Dynamo applied the technique to a

replicated key-value store and introduced virtual nodes, several positions per physical machine, to even out the load [21].

Hashing keys directly to machines has a drawback when the state behind each key is large or when ownership must be transferred carefully: every membership change touches many independent keys. Systems with expensive ownership transfer therefore introduce a fixed intermediate layer. Slicer, at Google, maps keys to slices and assigns slices to tasks, rebalancing the assignment from observed load [22]. Shard Manager at Meta generalises this into a platform that places and migrates shards for hundreds of applications, and reports that a fixed shard count with movable assignment is the dominant pattern in practice [6]. Borg schedules the processes that host such shards [23].

Taken into the design. The engine uses the two-level scheme of Slicer and Shard Manager: a workflow hashes to one of a fixed number of shards for its whole life, and shards are assigned to nodes by a consistent-hash ring with virtual nodes. A change of membership moves ownership but no data, because all data is in the shared database.

2.3 Leases, Coordination and Fencing

Assigning a shard to a node is only half of the problem; the system must also ensure that a node that has lost a shard stops writing to it. Gray and Cheriton introduced leases, time-bounded grants of authority that expire unless renewed, so that a crashed holder cannot block others forever [8]. A lease is safe only under assumptions about clock drift and process pauses. A holder that is paused, for example by garbage collection or by a virtual machine being descheduled, may resume after its lease has expired and act on authority it no longer has. Kleppmann describes this failure and the standard remedy: a *fencing token*, a number that increases with every grant, which the storage layer checks on every write [9].

Production systems usually obtain leases from a coordination service. Chubby provides coarse-grained locks with sequencers that serve as fencing tokens [24], and ZooKeeper provides the primitives from which leader election and locks are built [7]. Both rest on a consensus protocol such as Paxos or Raft [10]. The impossibility result of Fischer, Lynch and Paterson shows why some timing assumption is unavoidable: in a purely

asynchronous system no protocol can both tolerate a crash and always terminate [25]. Leases embody that assumption explicitly.

Taken into the design. A coordination service is not required if the store that holds the protected data can itself perform a linearizable compare-and-swap. PostgreSQL can, through row locks. The engine therefore keeps one lease row per shard and makes every write transaction check the fencing token under `SELECT ... FOR UPDATE` [26]. The token check happens in the same transaction as the write, which is the property the literature identifies as essential.

2.4 Long-Running Transactions and Sagas

Garcia-Molina and Salem defined a saga as a long-lived transaction written as a sequence of shorter transactions, each paired with a compensating transaction that semantically undoes it [27]. If the saga fails part way, the compensations for the completed steps are run in reverse order. Helland later argued that applications which span services cannot rely on distributed transactions at all, and must be built from idempotent operations on individually addressed entities [28]. Recent work on microservices has examined the isolation anomalies that sagas permit and proposed remedies such as semantic locks and versioning [29].

Two points from this literature matter here. A compensation is an action with its own side effects and can itself fail, so it must be retried and deduplicated in the same way as a forward step. The order and completeness of compensation must also survive a crash in the middle of rolling back.

Taken into the design. Compensations are ordinary activities, registered as each forward step completes and run in reverse order. Because they are activities, their scheduling and completion are recorded in the event history, and a crash during rollback is recovered by replay like any other.

2.5 Failure Modes of Agent Systems

Agents built on language models are a new class of long-running workflow. An agent plans, calls tools that act on the world, observes the results and plans again. Cemri

et al. analysed traces from several multi-agent frameworks and organised the observed failures into a taxonomy; step repetition and failure to recognise that a task was already complete appear among the recurring categories [11]. Microsoft's AI Red Team reached a similar list from the security side, including repeated or unintended actions after an interruption [12].

The relevance to durable execution is specific. A conventional engine records the arguments of each step when it is scheduled, so an engine-level retry of that step uses identical arguments. An agent that crashes between two planning steps, however, will re-run the second planning step on recovery, and at a non-zero sampling temperature it may describe the same action in different words. The engine sees a new step with new arguments.

2.6 Semantic Similarity and Caching

Sentence-BERT made it practical to compare the meaning of two sentences by the cosine similarity of fixed-size embeddings [30], and MiniLM showed that a small distilled model can produce such embeddings at a fraction of the cost of the original [31]. Semantic caches for language-model services build on this. GPTCache returns a stored answer when a new prompt is close enough to a cached one [32]; MeanCache adapts the threshold to each user [33]; and vCache argues that a single global threshold is unreliable and learns per-entry thresholds with an error bound [34].

A semantic cache and a semantic deduplication check fail in opposite directions. A false hit in a cache returns a slightly wrong answer, which the user can retry. A false match in a deduplication check suppresses an action the user wanted, which is worse than missing a duplicate. The cache literature therefore informs the mechanism but not the policy.

Taken into the design. The engine uses a small sentence-embedding model and a cosine threshold, but adopts a conservative policy: semantic matching is opt-in per activity, candidates are restricted to completed activities of the same type in the same run, an uncertain band between two thresholds is escalated rather than decided, and every verdict is recorded in the history.

2.7 Comparison of Related Systems

Table 2.1 compares the systems closest to this work on the properties that the engine is designed to combine.

Table 2.1: Comparison of related systems

System	Infrastructure	Partitioned	Ownership safety	Semantic dedup
Temporal [1]	4 services + DB (+ search)	Yes	Range IDs per shard	No
Durable Functions [3, 16]	Cloud storage or Netherite	Yes	Partition leases	No
Beldi [18]	Serverless + key-value store	By key	Logged conditional writes	No
Boki [19]	Custom shared log	Yes	Log sequencer	No
DBOS [4]	PostgreSQL	No	Database transactions	No
Shard Manager [6]	ZooKeeper + control plane	Yes	Coordination service	Not applicable
GPTCache [32]	Vector store	Not applicable	Not applicable	Cache only
This work	**PostgreSQL**	**Yes**	**Fencing token in each write**	**Journalled verdict**

2.8 Research Gap

The survey leaves three gaps. First, the engines that partition workflows across machines do so with several cooperating services or a coordination service, while the engine that needs only a database does not partition. No system in the survey provides partitioned, fenced ownership using only the database that already stores the workflow state. Second, fencing is well understood in principle, but published descriptions rarely report how often a stale owner actually attempts a write in a real deployment, or how long ownership takes to move after a failure. Third, every engine deduplicates steps by exact comparison of arguments, which fails for an agent that re-plans after a crash, and the semantic caching work that could detect such repeats is designed for a setting in which a false match is cheap. This project addresses all three.

CHAPTER - 3

REQUIREMENTS AND ANALYSIS

This chapter states what the engine must do and the conditions under which it must do it. Functional requirements are numbered FR-1 to FR-12 and non-functional requirements NFR-1 to NFR-8, so that later chapters can refer to them. The chapter ends with the analysis that led to the chosen approach.

3.1 Functional Requirements

Table 3.1 lists the functional requirements. They follow from the objectives in Table 1.2.

3.2 Non-Functional Requirements

3.3 Software and Hardware Requirements

The engine is written in Go 1.25 and uses the `pgx` driver for PostgreSQL. PostgreSQL 16 with the `pgvector` extension stores all state, including the 384-dimensional intent embeddings used for semantic deduplication. Embeddings are produced by an `all-MiniLM-L6-v2` model served over HTTP. Metrics are exported in Prometheus format. Development and local testing used Docker Compose; the cloud evaluation used Amazon EKS with Terraform for provisioning. Table 3.3 gives the hardware used in each setting.

3.4 Assumptions and Constraints

The design rests on the following assumptions.

1. PostgreSQL provides serializable row locks and durable commits. The engine does not defend against a database that loses committed data.
2. Clocks on engine nodes may drift by a small amount relative to the lease period. Lease expiry is computed with the database clock, so drift between engine nodes does not affect safety; fencing protects against pauses longer than the lease.

Table 3.1: Functional requirements

ID	Requirement
FR-1	A client can start a workflow of a registered type with an identifier and an input. Starting the same identifier twice returns the existing run rather than creating a second one.
FR-2	Every state change of a run is appended to its history as an immutable event with a gap-free sequence number.
FR-3	A run interrupted by a crash at any point resumes from its history and reaches the same result it would have reached without the crash.
FR-4	Activities are executed by workers through a durable task queue, with a configurable retry policy and per-task lock timeout.
FR-5	Each activity receives a deduplication key that is identical on every attempt and every replay.
FR-6	Workflows can wait on durable timers and on named signals sent by clients; both survive restarts.
FR-7	A workflow can register compensations; on failure they run in reverse order of registration, and rollback survives a crash.
FR-8	Workflows are partitioned into a fixed number of shards, each owned by at most one node at a time.
FR-9	When a node fails, its shards are claimed by surviving nodes and their runs continue without loss of progress.
FR-10	A write by a node that no longer owns the shard is rejected.
FR-11	An activity may opt in to semantic deduplication. A request whose meaning matches a completed activity of the same type in the same run returns that activity's recorded result instead of executing again.
FR-12	Clients can query the status and history of a run, and operators can read metrics for throughput, latency, ownership and errors.

Table 3.2: Non-functional requirements

ID	Property	Requirement
NFR-1	Correctness	No completed activity is executed again by the engine because of a crash, retry or ownership change, apart from the unavoidable window described in Section 4.2.4.
NFR-2	Recovery	After a node is killed, its shards are claimed within one lease period plus one renewal interval.
NFR-3	Dependencies	PostgreSQL is the only infrastructure dependency. The embedding service is needed only by activities that use semantic deduplication, and never during replay.
NFR-4	Scalability	Throughput increases when nodes or workers are added, until the database becomes the limit.
NFR-5	Efficiency	An idle cluster places a small, bounded load on the database.
NFR-6	Observability	Every node exports Prometheus metrics and structured logs; every semantic verdict is recorded with its inputs.
NFR-7	Portability	The engine runs as a single static binary in a container on Kubernetes or directly on a host.
NFR-8	Testability	Fault injection hooks allow crashes, pauses and database outages to be triggered at named points in tests.

Table 3.3: Hardware used for development and evaluation

Role	Instance	Resources
Development	Laptop / cloud container	4–8 vCPU, 16 GB, Docker
Engine nodes (cloud)	3 × m7i-flex.large	2 vCPU, 8 GB each
Load harness (cloud)	1 × c7i-flex.large	2 vCPU, 4 GB
Database (cloud)	1 × m7i-flex.large	2 vCPU, 8 GB, gp3 volume

3. Workflow code is deterministic. The engine provides deterministic replacements for time, random numbers and unique identifiers, and detects some violations at replay time, but cannot prove determinism in general.
4. External services that are called by activities accept an idempotency key, or their operations are naturally idempotent.

Two constraints were fixed at the start. The engine must not introduce any infrastructure dependency beyond PostgreSQL, and all cloud resources must fit the free-tier and low-cost instance types available to the project, which ruled out larger database hosts and multi-zone deployments.

3.5 Analysis of the Chosen Approach

Three decisions shape the rest of the design, and each was taken by comparing alternatives against the requirements.

State: event history or checkpoints. A checkpointing design stores the current state of each run and overwrites it after each step. It is simple, but it records what the state is and not how it was reached, so a crash between a side effect and the checkpoint leaves no trace of the side effect, and debugging requires extra logging. An event history records every decision; the current state is a function of the history, and the history is also the audit trail. The cost is replay time for long histories. Event sourcing was chosen because FR-2, FR-3 and FR-11 all depend on the history being the single source of truth.

Ownership: coordination service or database rows. A coordination service such as ZooKeeper or etcd provides leases and watches, but violates NFR-3 and adds a second system whose view of ownership can disagree with the database. More importantly, a lease held in one system does not by itself prevent a stale write to another: the token must be checked where the data is written. Keeping the lease and the fencing token in the same database as the workflow state lets a single transaction check ownership and write the data, which satisfies FR-10 directly.

Task delivery: message broker or database queue. A broker would offer higher raw throughput, but an activity's completion must be recorded in the history atomically with the removal of its task; with a separate broker this requires an outbox and a relay. A queue table polled with `SELECT ... FOR UPDATE SKIP LOCKED` gives atomic hand-off within one transaction. The cost, measured in Chapter 7, is polling load on the database when the cluster is idle and lock contention at high load.

The price of these decisions is that the database is both the availability floor and the throughput ceiling of the system. The evaluation in Chapter 7 was designed to locate that ceiling and to show what happens beyond it.

CHAPTER - 4

SYSTEM DESIGN

This chapter describes the design of the engine in two parts. The high-level design (Section 4.1) presents the architecture, the separation of workflow and activity code, the mapping of workflows to shards and nodes, and the ownership protocol. The low-level design (Section 4.2) gives the data model, the workflow task loop, the crash-window analysis, sagas and the semantic deduplication path.

4.1 High-Level Design

4.1.1 Governing principle

One rule governs the design: *the event history is the only state of a workflow run, and every decision is committed to it before it is acted on*. A node holds caches, worker goroutines and open connections, but nothing that would be lost if the process were killed. Recovery therefore never needs to reconstruct a node's memory; it needs only to replay a run's history, on any node that currently owns the run's shard.

4.1.2 Architecture

Fig. 4.1 shows the components. Clients reach any engine node through an HTTP API to start runs, send signals and read status. Each node is one Go process with four parts: the API server, a lease manager that acquires and renews shard leases, a pool of workflow workers that execute workflow code, and a pool of activity workers that execute side effects. All nodes run the same binary; any of them can own any shard. PostgreSQL holds the event log, the task queue, timers, leases and intent embeddings. There is no other shared component.

4.1.3 Separation of workflow and activity

Workflow code and activity code have opposite requirements, and the engine treats them differently. Workflow code decides what to do next. It is re-executed on ev-

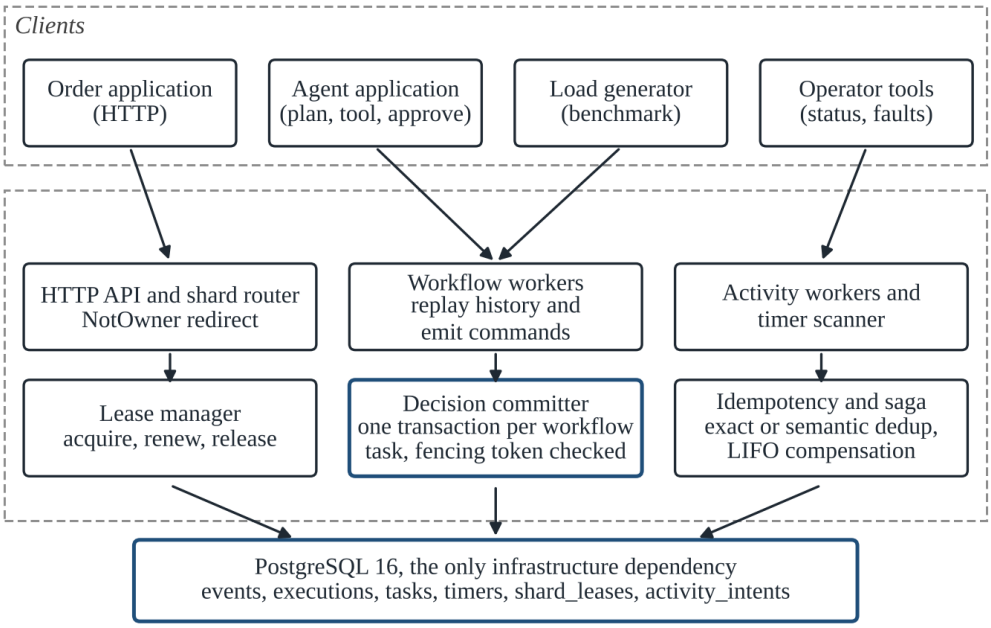


Fig. 4.1: System architecture. The engine node is one Go process, replicated N times; PostgreSQL is the only shared component.

ery recovery, so it must be deterministic and must perform no input or output; it calls `ExecuteActivity`, `Sleep`, `WaitForSignal` and `SideEffect` on a context object, each of which either returns a result recorded in the history or suspends the workflow until one is available. Activity code performs the side effects. It may be non-deterministic, may fail and may take a long time; it runs at least once and its result is recorded when it completes.

A workflow function in the engine's Go SDK has the following shape.

Listing 4.1: A workflow with two activities and a saga compensation

```
func Order(ctx workflow.Context, in OrderInput) (Receipt, error) {
    s := saga.New(in.OrderID, runner(ctx))
    var pay Payment
    if err := ctx.ExecuteActivity("charge_card", in.Card).Get(&pay); err !=
        nil {
        return Receipt{}, err
    }
    s.AddCompensation("refund_card", pay.ID)   // only after success
    var res Reservation
```

```
    if err := ctx.ExecuteActivity("reserve_stock", in.Items).Get(&res); err
        != nil {
        return Receipt{}, s.Compensate()        // runs refund_card
    }
    return Receipt{Payment: pay.ID, Reservation: res.ID}, nil
}
```

4.1.4 Two-level mapping of workflows to nodes

A workflow identifier is hashed to one of S shards, with $S = 16$ by default, and the shard never changes for the life of the run. Shards are assigned to nodes by a consistent-hash ring with 150 virtual nodes per engine node (Fig. 4.2). The ring expresses a preference only: a node reaches for the shards the ring assigns to it, and holding a shard is decided by its lease. Because all data lives in the shared database, moving a shard between nodes moves no data; the new owner simply begins to poll that shard's task rows.

Two rules make the preference converge after failures. A node may adopt a shard that the ring does not assign to it if the shard's lease has been expired for more than twice the lease period, which is taken as evidence that the owner is not coming back. A node that holds a shard the ring assigns to another node returns it as soon as that node answers its health check. The first rule prevents a dead node's shards from stalling forever; the second keeps the load even after the node recovers. The delay in the first rule avoids needless handovers after a short pause or a rolling restart, and its cost appears in the freeze results of Chapter 7.

4.1.5 Leases and the fencing invariant

Each shard has one row in `shard_leases` holding the owner, an expiry time and a fencing token. A node acquires a free or expired lease with a single conditional update that increments the token and sets the expiry to the database's `now()` plus the lease period of 10 s. It renews every 3 s, so a node has three attempts before its lease lapses. Expiry is computed on the database clock, so clock skew between engine nodes does not matter.

A lease alone is not safe. A node that is paused for longer than the lease period resumes believing it still owns its shards, while another node has already taken them over. The engine therefore enforces the following invariant at the point of writing:

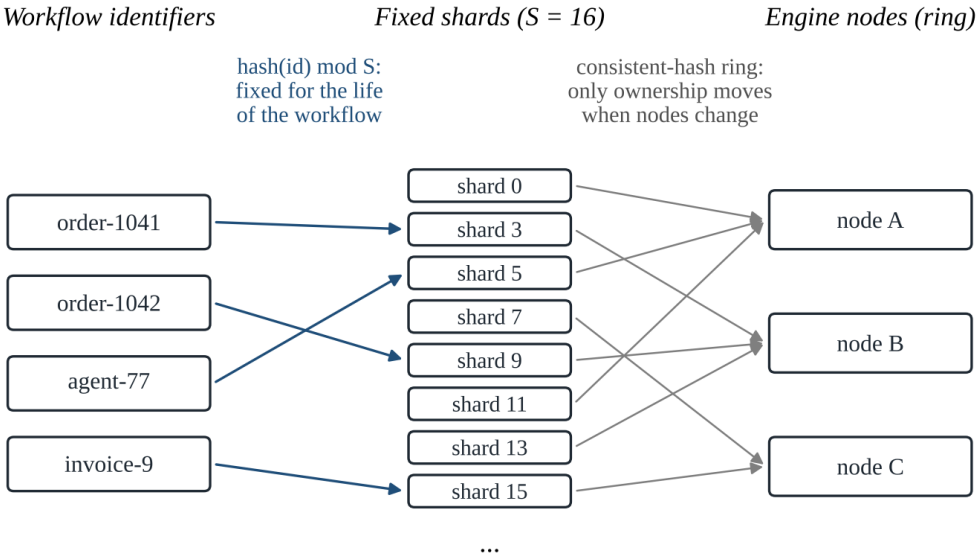


Fig. 4.2: Two-level mapping. Workflows hash to a fixed set of shards; shards are placed on nodes by a consistent-hash ring.

> *Every transaction that changes the state of a shard first reads that shard's fencing token with* `SELECT ... FOR UPDATE`*, and commits only if the token equals the one the writer was granted.*

The row lock is held until the transaction ends, so ownership cannot change between the check and the commit; a competing acquire waits for the lock and then increments the token. A stale owner's transaction finds a higher token and rolls back, and none of its writes become visible. Fig. 4.3 shows the sequence as it occurred on the cloud test bed when an engine process was frozen for 35 s.

The token check is also the reason a separate coordination service is not needed. A lease obtained from ZooKeeper or etcd would still require a token check where the data is written; with the lease in the same database, the check and the write are one transaction.

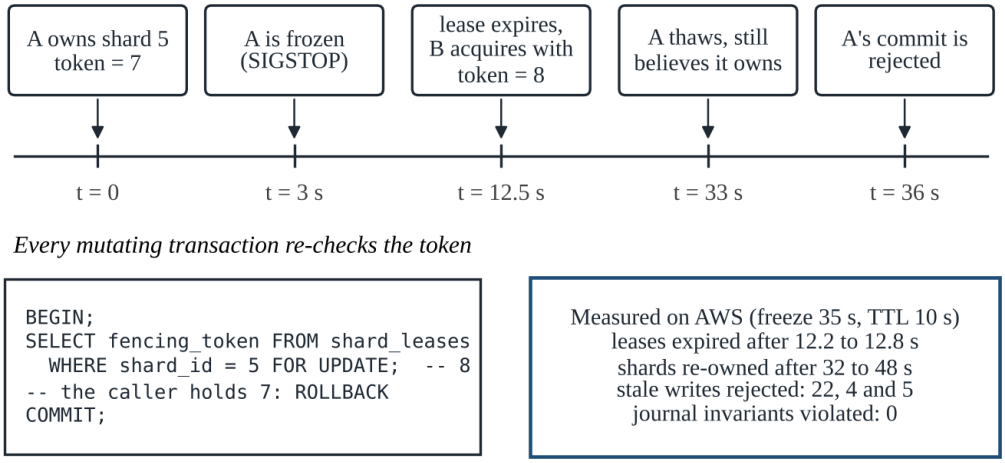


Fig. 4.3: Lease expiry and fencing of a stale owner, with the values measured in the freeze trials.

4.2 Low-Level Design

4.2.1 Data model

Table 4.1 lists the six tables. Every table except `shard_leases` carries the shard identifier as the first column of its primary key or its main index, so that each node's queries touch only the rows of the shards it owns.

Table 4.1: Persistent data model

Table	Key	Contents
`events`	(shard, workflow, run, seq)	Event type and JSON payload; append-only, gap-free `seq`
`executions`	(shard, workflow)	Run identifier, type, status and the last sequence number
`tasks`	(shard, id)	Workflow and activity tasks with visibility time, lock expiry, lock token and attempt count
`timers`	(shard, workflow, run, seq)	Fire time of each durable timer
`shard_leases`	shard	Owner, fencing token and expiry
`activity_intents`	(shard, workflow, run, seq)	Canonical intent text and its 384-dimensional embedding

The primary key of `events` is a second line of defence: even if two writers passed the fencing check, they could not both append the same sequence number.

4.2.2 The workflow task loop

A workflow makes progress only when a workflow task for it is processed. Such a task is created when the run starts and whenever something the run is waiting for happens: an activity completes or fails, a timer fires or a signal arrives. Fig. 4.4 shows how a worker processes one task.

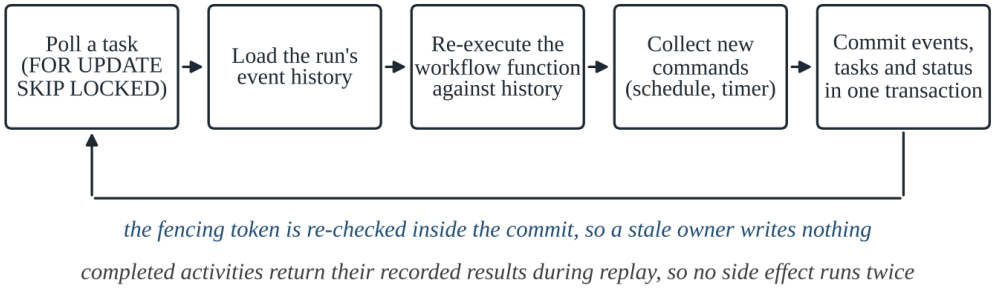


Fig. 4.4: The workflow task loop. All writes of one decision commit in a single fenced transaction.

The worker loads the run's history and executes the workflow function from the beginning. Each call on the workflow context is numbered by its position in program order, $k = 0, 1, 2, \ldots$. If the history holds a result for call k, the call returns that result immediately. The first call with no recorded result ends the replay: the worker collects the new commands the workflow has issued (schedule an activity, start a timer, complete the run) and commits them as one *decision*. The commit performs, in order: the fencing check; admission of the run for a start, using the `executions` primary key as the start-idempotency check; appending the new events; inserting the new tasks, timers and intents; updating the execution status; and deleting the processed task. Either all of these become visible or none do.

Using the program-order index k as the join key between code and history is what makes replay exact. A workflow that issues the same calls in the same order matches its history call by call. A workflow whose code has been changed in a way that alters the order is detected on replay, because the command at position k no longer matches the recorded event type, and the run is failed with a non-determinism error instead of continuing with a corrupted history.

Workflow code is run on a cooperative scheduler. Each workflow coroutine runs until it blocks on a future, and the scheduler then runs the next ready one, so that a workflow that starts several activities in parallel is still executed in a deterministic order. Time, random numbers and unique identifiers are obtained from the context, which returns recorded values during replay.

4.2.3 Task queue, timers and signals

Workers poll the `tasks` table with `SELECT ... FOR UPDATE SKIP LOCKED LIMIT 1`, restricted to the shards the node owns and to rows whose visibility time has passed and whose lock has expired. The selected row is locked by setting `locked_until` and a new `lock_token`. `SKIP LOCKED` lets many workers poll the same shard without waiting on each other. A long-running activity extends its lock with a heartbeat; if the worker dies, the lock expires after 30 s and the task is delivered again.

Timers are rows with a fire time. A timer loop on the owning node selects due timers, appends a `TimerFired` event and creates a workflow task, in one fenced transaction. A signal is appended directly to the target run's history as a `SignalReceived` event together with a workflow task, so a signal sent while the run's shard is moving between nodes is not lost.

4.2.4 Retries and the crash-window analysis

When an activity fails, the workflow worker applies the activity's retry policy: an initial interval of 1 s, a backoff coefficient of 2, a maximum interval of 60 s and at most five attempts by default, with a list of error types that are not retried. The retry is a new task with a later visibility time; the arguments are those recorded when the activity was first scheduled.

Table 4.2 considers each point at which a crash can occur during one activity. In three of the four windows recovery is safe without help from the activity. In window C the external effect has happened but the engine has no record of it, and the activity will be called again.

Window C cannot be closed by the engine, because the external service and the engine's database are separate systems with no common transaction. The engine provides the means for the external service to close it: every activity receives a deduplication key

Table 4.2: Crash windows for one activity execution

	Crash point	Recovery	Safe
A	Activity scheduled, not yet polled	Another worker polls the task; the activity runs once	Yes
B	Task polled, external call not made	The lock expires and the task is redelivered; the activity runs once	Yes
C	External call succeeded, completion not recorded	The lock expires and the task is redelivered; the external call is made again	No
D	Completion recorded, workflow task not yet processed	Replay resolves the activity from history; it is not run again	Yes

formed from the run identifier and the program-order index k, which is the same on every attempt and every replay. An activity that passes this key as the idempotency key of the external call turns a repeated call into a no-op at the service. The engine's guarantee is therefore at-least-once execution of activities and exactly-once recording of their results.

4.2.5 Saga compensation

Compensations are managed by a small saga object created inside the workflow (Listing 4.1). A compensation is added with `AddCompensation` only after its forward step has returned successfully, so a step that failed or never ran has nothing to undo. When a later step fails, `Compensate` runs the registered compensations as activities in reverse order of registration.

The stack of compensations is not written to the history. It is a pure function of the history, because on replay the workflow follows the same path, observes the same successes and registers the same compensations in the same order. Recording it would add an event per step and would create a second copy that could disagree with the first. Each compensation is scheduled and completed through the history like any other activity, with its own retries and deduplication key, so a crash in the middle of a rollback resumes at the first compensation that has not completed.

4.2.6 Semantic deduplication

The source of reworded duplicates is narrower than it may appear. An engine retry of a scheduled activity reuses the recorded arguments, so exact matching is enough. Reworded duplicates arise when an agent re-plans: replay resolves an earlier plan and its tool call from history, but a later planning step whose result was not yet recorded runs again and may express the same intent in different words.

The similarity check is itself non-deterministic, since it depends on an embedding model and on a threshold. If it were evaluated in the activity worker, it would be re-evaluated on replay and could give a different answer, and the history would diverge. The engine therefore evaluates it once, on the workflow path, and records the verdict as an event, in the same way as a `SideEffect`. Listing 4.2 gives the procedure.

Listing 4.2: Semantic deduplication of an activity at position k

```
first execution:
  intent     := canonical(type, normalised and sorted arguments)
  candidates := completed semantic activities of the same type in this run
  v          := embed(intent)                      // 384 dimensions
  (j, s)     := argmax over candidates of cosine(v, v_j)
  verdict    := DUPLICATE if s >= tau_high (0.90)
                DISTINCT  if s <  tau_low  (0.85)
                ESCALATE  otherwise
  append SemanticDedupEvaluated{k, verdict, j, s, thresholds, model}
  DUPLICATE: return the recorded result of activity j; schedule nothing
  DISTINCT:  schedule the activity as usual
  ESCALATE:  wait for an approval signal, or execute (per configuration)
replay:
  read SemanticDedupEvaluated{k} and apply the recorded verdict
  (no embedding call, no similarity computation)
```

Three properties follow. Replay needs no embedding service, so recovery does not depend on its availability. Every verdict is stored with its similarity, thresholds, matched activity and model name, so a suppressed action can be explained after the fact. The candidate set is restricted to completed activities of the same type in the same run, which keeps the comparison small and makes a false match across unrelated runs impossible. The intent embeddings are stored in `activity_intents` with the `pgvector` type.

4.2.7 Node runtime

At start-up a node applies the schema migrations, connects to PostgreSQL, builds the ring from the configured list of cluster nodes and starts the lease manager. For each shard it acquires, it starts workflow, activity and timer loops for that shard; when it loses a shard it stops them at once. The fencing check would reject their writes in any case, but stopping them avoids polling load on a shard the node no longer owns. On a graceful shutdown the node stops taking new tasks, waits for in-flight tasks within the shutdown grace period and releases its leases, so that the next owner can take over without waiting for expiry. The defaults are four workflow workers and eight activity workers per node; Chapter 7 measures the effect of changing them.

CHAPTER - 5

IMPLEMENTATION

The engine was implemented in Go 1.25 as a single binary, `engined`, of about 43,000 lines of non-test code, together with roughly 23,000 lines of tests. This chapter describes the structure of the code and the parts of the implementation where the details matter for correctness or performance. The design that the code realises was given in Chapter 4 and is not repeated.

5.1 Code Organisation

Table 5.1 lists the main packages. Dependencies point downwards: `core` defines the event types, identifiers and interfaces and imports nothing from the engine, and the storage packages depend only on `core` and the PostgreSQL driver. The runtime package wires the pieces together, and `cmd/engined` builds the process from configuration.

Table 5.1: Main packages of the engine

Package	Lines	Responsibility
`core`	625	Event types, identifiers, decision and store interfaces
`eventlog`	1,005	Reading histories; the fenced decision commit
`taskq`	1,002	Durable task queue: poll, lock, heartbeat, complete, retry
`lease`	789	Lease store and the ownership state machine
`shard`	275	Workflow-to-shard hash and the consistent-hash ring
`timers`	264	Durable timers and the due-timer scanner
`workflow`	1,644	Workflow context, replay by program order, cooperative dispatcher
`activity`	729	Activity registry, retry policy, deduplication keys
`saga`	202	Compensation stack
`idem`	3,454	Exact and semantic deduplication, embedder client, hybrid comparator
`runtime`	2,954	Workflow, activity and timer workers; per-shard worker pools
`api`	1,959	HTTP API, error mapping, request routing
`obs`	444	Prometheus metrics and structured logging
`faultinj`	364	Named crash points for tests

5.2 The Fenced Commit

All writes produced by one workflow task are made by a single function, `CommitDecision`, in one transaction. Its first statement is

Listing 5.1: Fencing check at the start of every commit

```
SELECT fencing_token FROM shard_leases WHERE shard_id = $1 FOR UPDATE
```

and the function returns a `NotOwner` error, after rolling back, if the value differs from the token held by the caller. The remaining steps follow the order given in Section 4.2.2. Placing the start of a new run after the fencing check was a deliberate change made during development: in an earlier version the insert into `executions` for a start was performed outside the fenced transaction, so a node that had lost a shard could still admit new runs to it. The same pattern, a token check under a row lock in the writing transaction, is used by the timer scanner and by the signal path.

5.3 Task Queue

Each worker polls one owned shard at a time with a query that selects one ready row and locks it in the same statement:

Listing 5.2: Task poll (simplified)

```
WITH pick AS (
  SELECT shard_id, task_id FROM tasks
   WHERE shard_id = $1 AND task_type = $2 AND ready_at <= now()
   ORDER BY shard_id, task_type, ready_at
   FOR UPDATE SKIP LOCKED LIMIT 1)
UPDATE tasks t
   SET locked_until = now() + $3::interval,
       lock_token = gen_random_uuid(), attempt = t.attempt + 1
  FROM pick p
 WHERE t.shard_id = p.shard_id AND t.task_id = p.task_id
RETURNING t.*
```

Here `ready_at` stands for the later of the visibility time and the lock expiry. A partial index on ready tasks, added in the second migration, keeps the inner query to an index scan. Completion and failure are accepted only with the lock token issued by the poll, so

a worker whose lock has expired and whose task has been given to another worker cannot complete it. Idle workers back off between polls; Chapter 7 shows that this polling is the dominant database load on an idle cluster and that it sets part of the latency floor.

5.4 Lease Manager

The ownership policy is written as two pure functions. `Decide` takes the remembered state of one shard and the current time and returns one action: acquire, renew, declare lost or do nothing. `Apply` takes the outcome of the database call and returns the new state. Neither function touches the database or a clock, so all combinations of late renewals, failed queries and expired leases are covered by table-driven unit tests that run in microseconds. Two rules in these functions matter for safety. Expiry is checked before renewal, so a node never renews a lease that has already lapsed. A renewal that fails because the database is unreachable does not advance the time of the last successful renewal, so if the failure persists the node gives up the shard on its own. The loop that calls these functions runs every renewal interval and calls back into the runtime to start or stop the shard's workers.

5.5 Workflow Execution

Each workflow task runs the workflow function in a goroutine controlled by a dispatcher. When the function calls `Get` on a future that has no recorded result, the goroutine hands control back and is then discarded; the run will be re-executed from the start on its next task. This keeps the runtime free of long-lived per-run goroutines, so a run sleeping for a week costs one row in `timers` and nothing in memory. The context counts calls in program order and, on each call, compares the requested command with the event recorded at that position; a mismatch fails the run with a non-determinism error that names the position, the recorded command and the requested one.

5.6 Semantic and Hybrid Deduplication

The semantic checker builds a canonical intent string from the activity type and its normalised, sorted arguments, obtains an embedding from the embedder service, and compares it by cosine similarity with the stored embeddings of completed activities of the same type in the run. The verdict and its inputs are journalled as a

`SemanticDedupEvaluated` event before any command is issued. Lookups and inserts of embeddings use the `pgvector` type in `activity_intents`.

The evaluation in Chapter 7 showed that a general sentence encoder answers a different question from the one deduplication asks. It judges two requests for the same route on different days as very similar, which they are, though they are two different bookings. An opt-in *hybrid* comparator was therefore added. The activity declares its deciding fields, for example the origin, destination and date of a flight; these are normalised ("BLR" and "Bangalore" map to one code, "Friday AM" to a date and time window) and compared exactly, and similarity is applied only to the remaining free text. The verdict is journalled in the same way, so the hybrid comparator changes the decision but not the replay mechanism.

5.7 API, Routing and Observability

The HTTP API (Table 5.2) is served by every node. A request for a run on a shard the node does not own is answered with HTTP 409 `NotOwner` and, when known, the name of the current owner, and clients retry against that node. Every node exports Prometheus metrics for task latencies, commits, lease transitions, fencing rejections, deduplication verdicts and connection-pool use, and writes structured JSON logs. The fencing-rejection counter is the one used in Chapter 7 to count stale writes.

Table 5.2: HTTP API

Endpoint	Purpose
`POST /v1/workflows`	Start a run (idempotent on identifier)
`GET /v1/workflows/{id}`	Status of a run
`GET /v1/workflows/{id}/history`	Event history of a run
`POST /v1/workflows/{id}/signals`	Send a named signal
`POST /v1/workflows/{id}/cancel`	Request cancellation
`GET /v1/shards`	Current lease table
`GET /v1/dedup/verdicts`	Recorded semantic verdicts
`GET /healthz`, `GET /metrics`	Liveness and metrics

5.8 Demonstration Applications

Two applications exercise the engine. The first is a food-delivery order written as a saga: payment, restaurant confirmation, driver assignment and notification, each with a compensation, and a durable timer for the preparation time. Every step has a natural exact key such as the order identifier, so it uses no semantic deduplication. It is the workload used for all load tests in Chapter 7. The second is a travel agent whose planner proposes flight and hotel bookings. It can run with a recorded fixture or a live language model, and it reproduces the double-booking case of Section 1.2.3 when a crash is injected between two planning steps.

5.9 Deployment

The binary is built into a small container image. Locally, Docker Compose starts three engine nodes, PostgreSQL with `pgvector`, the embedder, Prometheus and Grafana. On Kubernetes the engine runs as a StatefulSet, so that each pod has a stable name that is also its node identifier on the ring; pod anti-affinity places one engine per node. The AWS environment of Chapter 6 is created by Terraform and contains no component other than those listed here: no message broker, cache or coordination service.

CHAPTER - 6

TESTING AND EVALUATION METHOD

The engine was tested at four levels: unit tests of individual packages, integration tests against a real PostgreSQL instance, fault-injection tests at named crash points, and end-to-end tests on a local three-node cluster. It was then evaluated on a cluster in Amazon Web Services. This chapter describes the tests and the local results, and then the cloud test bed and the measurement procedure whose results are reported in Chapter 7.

6.1 Unit and Integration Tests

The suite contains 591 test functions, which expand to 1,223 test cases when table-driven subtests are counted. Table 6.1 gives their distribution. Tests that need a database start a disposable PostgreSQL schema through the `testdb` helper, so the storage code is never tested against a mock. Three groups of tests carry most of the weight for correctness.

1. *Ownership.* The pure `Decide` and `Apply` functions of the lease manager are tested over all combinations of lease state, renewal outcome and elapsed time. Integration tests run two lease managers against one database and check that a shard never has two owners whose tokens are both accepted.
2. *Fencing.* A commit is attempted with a stale token for each kind of write (start, decision, timer, signal) and the test checks that nothing from the transaction is visible afterwards.
3. *Replay.* Workflows are run, interrupted at each call position and re-executed against the stored history; the result must be identical and no activity may be scheduled twice. Changed workflow code must be reported as a non-determinism error.

Table 6.1: Distribution of test functions by package

Package	Tests	Package	Tests
`runtime`	70	`eventlog`	20
`api`	59	`config`	18
`workflow`	33	`faultinj`	15
`taskq`	32	`timers`	14
`lease`	31	`obs`	13
`activity`	25	`saga`	10
`shard`	21	others	230
Total test functions (test cases including subtests)			591 (1,223)

6.2 Fault Injection at Crash Points

The `faultinj` package defines six named crash points, armed by an environment variable: `BeforeStartCommit`, `BeforeDecisionCommit`, `BeforeResultJournal`, `AfterActivityJournaled`, `BeforeNextPlanCompletes` and `AfterLeaseRenew`. When the armed point is reached, the code returns an error at that boundary and the transaction in progress is abandoned. Three of the points were exercised on a local Kubernetes cluster by a script that arms the point, restarts the engines, drives a workflow through the boundary and inspects the database. A start abandoned at `BeforeStartCommit` left no execution row and was admitted cleanly on retry. A decision abandoned at `BeforeDecisionCommit` in the middle of a run, with three activities already recorded and a timer pending, was retried; the run completed with four scheduled and four completed activities and no duplicate event. A crash at `AfterActivityJournaled` likewise produced no duplicate (run, sequence number) or program-order position. The remaining points are covered by unit tests. These seams test the boundary of durable state; loss of a whole process was tested separately by killing processes, both locally and on the cloud test bed.

6.3 Local Cluster Tests

A Docker Compose cluster of three engine nodes was used to test behaviour that needs real processes. The most direct test is the duplicate audit: 1,000 delivery workflows were started at a concurrency of 64 while one node was killed after 8 s and a second 12 s later, and both were restarted while the batch was running. 763 runs completed.

The remaining 237 starts were refused with `NotOwner` while their shards had no owner; they were never accepted and so were never at risk. An audit of the 3,052 recorded activity completions found no duplicate (shard, workflow, run, k), and every accepted run ended in the `COMPLETED` state. A promotion drill with a streaming standby database was also carried out locally, but the cloud evaluation used a single database host and the drill is not discussed further.

The semantic deduplication mechanism was evaluated offline on a corpus of 213 pairs of flight-booking requests, 98 labelled duplicate and 115 distinct, of which 91 distinct pairs are lexically close to each other (for example, the same route on a different day). The pairs and their labels were produced by construction, not by human annotators, which is a limitation discussed with the results.

6.4 Cloud Test Bed

The cloud evaluation ran on the infrastructure shown in Fig. 6.1, in one availability zone of the `us-east-1` region. Three engine pods ran on three m7i-flex.large nodes, one pod per node, with the default configuration of 16 shards, four workflow workers and eight activity workers per shard. PostgreSQL 16 ran on a separate m7i-flex.large host with `synchronous_commit` on and no standby. The load generator ran on a separate, tainted node so that it competed with nothing. All commands were issued from an operator instance inside the VPC, reached through AWS Systems Manager, and every result file was copied to S3 with its command line, image digest and harness version.

6.5 Measurement Procedure

The load generator starts delivery workflows and records the time from the start request to completion. It runs in one of two modes, following the distinction drawn by Schroeder *et al.* [35]. In *closed-loop* mode, c clients each start a new workflow when their previous one completes, so the arrival rate falls when latency rises. In *open-loop* mode, workflows are started at a fixed rate regardless of completions, which is how independent users behave. By Little's law [36], a closed loop with c clients and latency W carries c/W workflows per second, so closed-loop throughput can look stable even when the system is overloaded. Both the median and the 99th percentile are reported, because

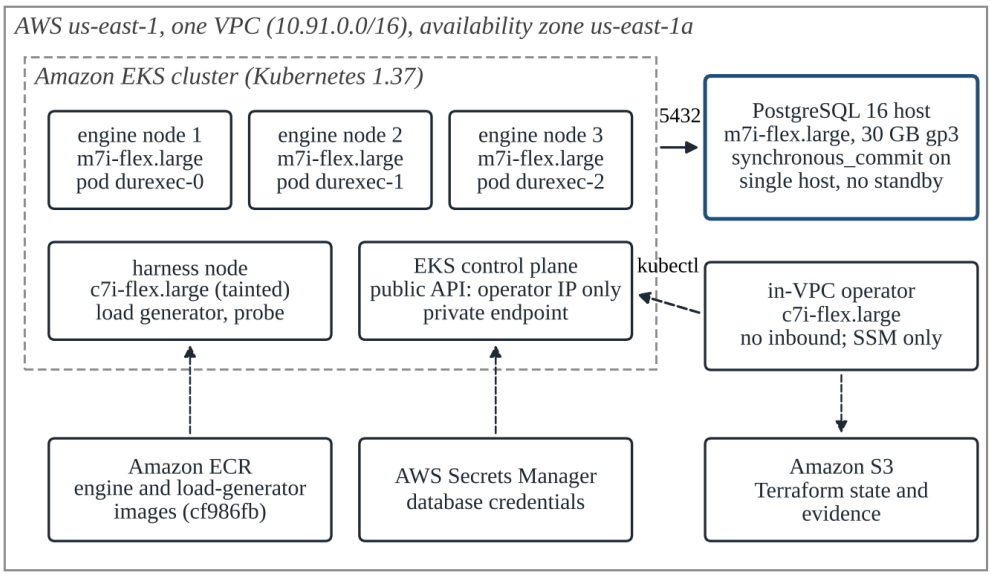


Fig. 6.1: Cloud test bed on AWS.

in a system of many cooperating parts it is the tail that determines the experience of users [37].

Each measurement point ran for 135 s, of which the first 15 s were discarded as warm-up; the calibration points ran for 435 s to collect enough samples. Table 6.2 lists the stages. Ladder stages screened a range of concurrencies with one trial each and then repeated the best two with three trials; the knee was chosen from the means of the repeats, not from the screen, because single trials near saturation were found to vary by up to 40%.

Table 6.2: Evaluation stages

Stage	Purpose	Procedure
B	Idle overhead	No workload; database commits per second for five configurations, 3 trials each
C	Latency floor	Closed loop at $c = 1$, and open loop at 1 wf/s, 3 trials each
D	Closed-loop capacity	c from 4 to 512; repeats at 128, 256, 512
E	Open-loop capacity	0.5 to 1.1 times the Stage D knee, 3 trials each
F	Worker sensitivity	Workers per shard 2+4 and 8+16 against the default 4+8
G	Shard sensitivity	32 and 64 shards at a constant 192 workers
H	Replica scaling	One and two engine replicas against three
Faults	Recovery	Six fault types under open-loop load (Section 6.6)

Every point was classified by the harness before it was accepted. A point was `RECORDED` if the generator itself was not a bottleneck (its CPU, lag and in-flight count were checked) and every started workflow completed. A point with failed or unfinished runs was marked for correctness review, and one in which the generator could not keep up was marked invalid and excluded. After every point, a set of SQL invariants was run over the journal: no duplicate (run, sequence number), no duplicate activity completion, no completion without a matching schedule, no gap in sequence numbers and no run with more than one terminal event. During each point the harness also sampled the CPU of every node, the database's commit rate, WAL throughput, wait events and lock waits, and the engines' connection-pool statistics.

6.6 Fault-Injection Procedure

Each fault trial ran an open-loop load of 65.6 workflows per second, half the closed-loop knee and a rate sustained without loss in Stage E, and injected one fault after the load had settled, in the manner of chaos engineering experiments on production systems [38]. Six faults were used, each with three trials on a different victim pod, except the database reboot, which was run once:

1. graceful deletion of an engine pod;
2. `SIGKILL` of the engine process;
3. freezing the engine process with `SIGSTOP` for 35 s, longer than the 10 s lease, then resuming it;
4. severing the engine's database connections;
5. draining the Kubernetes node that hosts an engine pod;
6. rebooting the database host.

The recovery of each trial was measured on the database clock, from the moment of injection, by polling the lease table and the event log (Fig. 6.2). t_{expire} is the time at which the victim's leases expired or moved; t_{claim} the time at which all of them were held again with a higher fencing token; t_{progress} the time of the first new event on those shards; and t_{ring} the time at which all pods were ready and no shard was unowned. The fencing-rejection counters of all engines were read before and after, and the journal

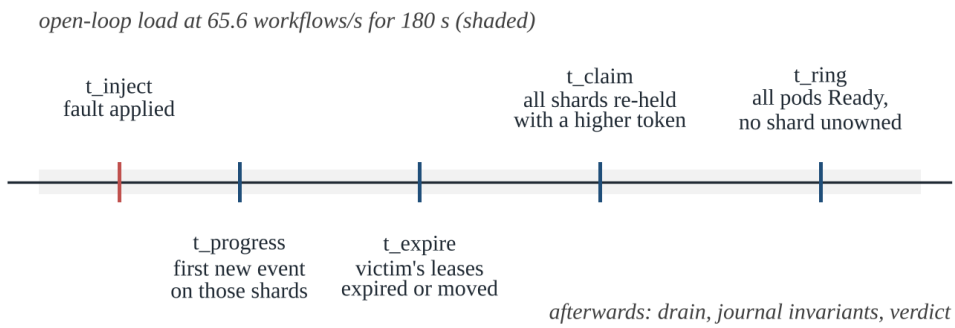


Fig. 6.2: Timing of a fault trial. All times are measured from injection on the PostgreSQL clock.

invariants were checked after the load had drained. A trial was valid only if the cluster was healthy before injection and the injection was confirmed to have happened; early attempts that failed these checks because of harness defects were kept in the results file as invalid and are not counted.

CHAPTER - 7

RESULTS AND DISCUSSION

This chapter reports the measurements made on the cloud test bed of Section 6.4 and the offline evaluation of semantic deduplication. Unless stated otherwise, the configuration is three engine replicas, 16 shards and four workflow and eight activity workers per shard (written r3/S16/w4x8), and the workload is the delivery workflow of Section 5.8. Values are means of three trials, with the range given where it is informative. All results describe this configuration on this hardware; Section 7.9 discusses how far they generalise.

7.1 Latency Floor

With one workflow in flight at a time, the median end-to-end latency was 2.21 s (p99 4.3–4.4 s), while the engine and database hosts were below 5% CPU. Offering 1 workflow per second in open loop raised the median by only 4–9%. The floor is therefore not set by any resource. Fig. 7.1 decomposes it using the timestamps in the event journal. Each activity takes about 200 ms from being scheduled to being recorded as complete, most of which is the simulated service time and the activity worker's poll interval. A fenced decision commit takes 33–38 ms. The largest single component is the lag of the 50 ms preparation timer, which fired a median of 580 ms late; the due-timer scanner runs once per second, so a timer waits on average half a second for the next scan.

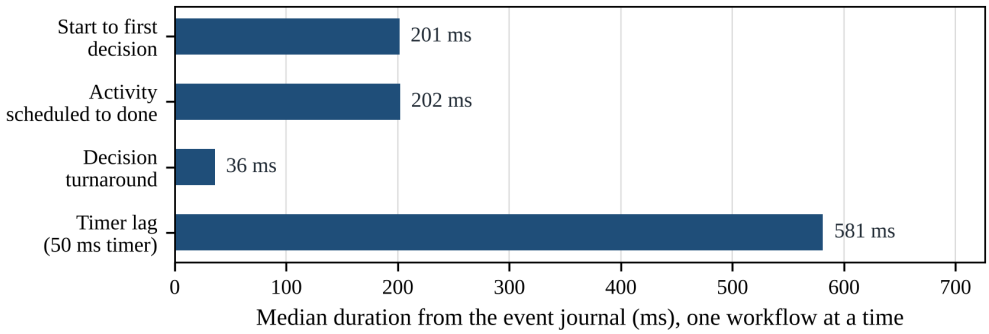


Fig. 7.1: Components of the latency floor at one workflow in flight (Stage C).

The floor is therefore a consequence of polling, and could be lowered by notifying workers of new tasks and timers instead of having them poll, for example with PostgreSQL's `LISTEN/NOTIFY`. Section 7.3 shows the same effect from the other side: under moderate load the median latency falls below the floor, because busy workers find work on their first poll and do not back off.

7.2 Idle Overhead

Because workers poll, an idle cluster is not silent. Fig. 7.2 shows the transactions committed per second by the database with no workload for five configurations. At the default configuration the idle rate was 251 commits/s at about 1% database CPU. The rate grows with the number of workers (165, 251 and 443 commits/s for 2+4, 4+8 and 8+16 workers per shard) and, at a fixed total of 192 workers, with the number of shards (251, 304 and 410 commits/s for 16, 32 and 64 shards), because each shard adds a timer scanner and a lease renewal. The trials were very repeatable, within 2 commits/s of each other.

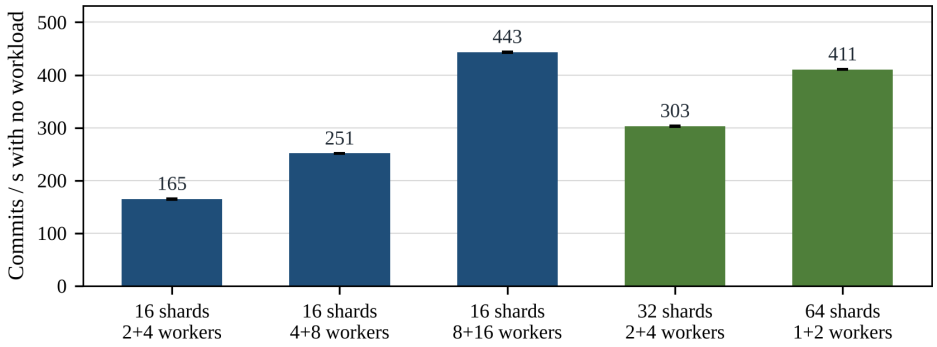


Fig. 7.2: Database commits per second with no workload (Stage B, three trials each).

The cost is small in CPU but it is not small in transactions, and it is paid on every host that runs the database whether or not any work is being done. It is the price of using a table as a queue without a notification channel.

7.3 Closed-Loop Capacity

Fig. 7.3 and Table 7.1 give throughput and latency as the number of concurrent workflows c was increased. Up to $c = 256$, throughput grew almost linearly with c and the median stayed near 2 s, which is what Little's law predicts for a system whose latency

does not change: $256/2.0\,\mathrm{s} \approx 128$ workflows/s against the 131 measured. In this range the system is limited by latency, not capacity.

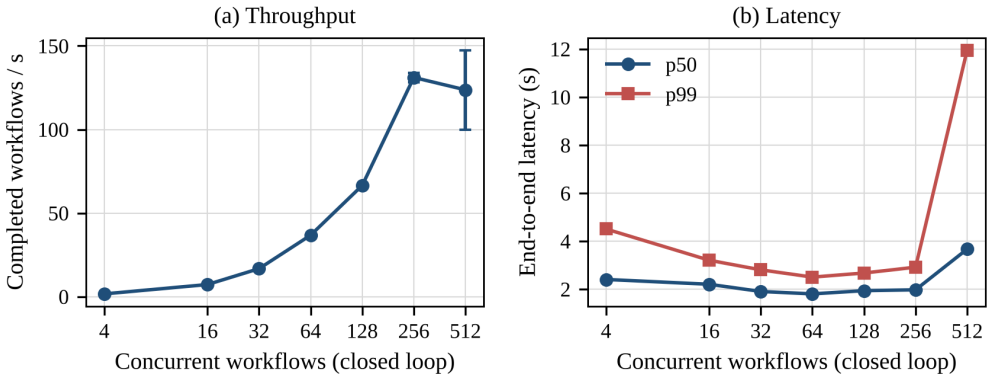


Fig. 7.3: Closed-loop throughput and latency against concurrency (Stage D).

Table 7.1: Closed-loop results (r3/S16/w4x8)

c	Workflows/s	p50 (s)	p99 (s)	DB CPU (%)	Commits/s
4	1.6	2.41	4.51	4	481
16	7.2	2.21	3.21	8	1,209
64	36.8	1.80	2.50	22	4,100
128	66.5	1.9–2.0	2.6–2.7	41–52	7,150
256	**131.1**	1.9–2.0	2.7–3.1	74–96	13,200
512	123.7	3.1–4.1	8.8–14.0	96	18,400–20,100

At $c = 512$ the mean throughput was lower than at 256 and varied widely between trials (151, 107 and 113 workflows/s), while p99 rose by three to five times. The first screening trial at 512 had suggested that throughput was still rising; the repeated trials showed that it was not. This is why the knee was taken from repeat means. The sustainable closed-loop capacity of this configuration is therefore about 131 workflows/s at $c = 256$.

Fig. 7.4 shows where the capacity goes. The database host reached 84% CPU on average at $c = 256$ and 96% at 512, while the busiest engine node stayed below 63% and the load generator below 30%. The engines' connection pools began to run empty at the same point, so the database and the pools saturate together. The wait profile of the database (Fig. 7.5) changes in a consistent way: at low load almost all sampled waits are for WAL flushes to disk, while from $c = 128$ onwards waits on the WAL write lock and on row locks dominate. The row-lock waits are consistent with the fencing check,

which serialises the commits of each shard on its lease row, although the sampling did not identify the rows.

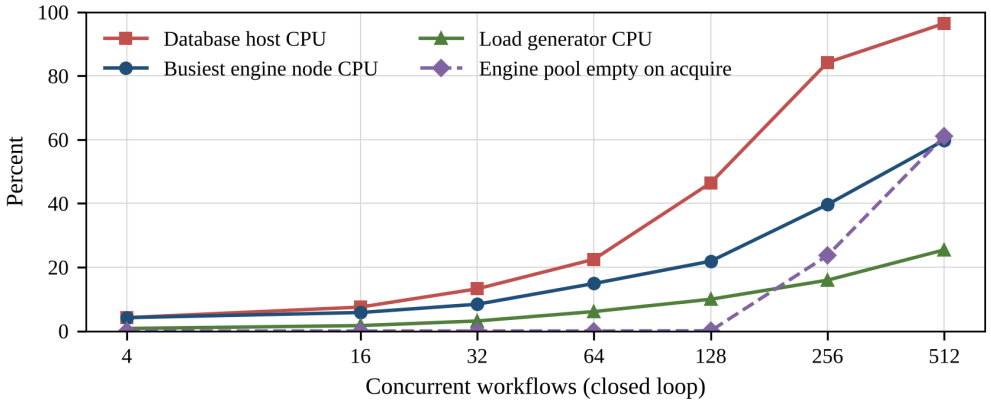


Fig. 7.4: Resource use against concurrency (Stage D).



Fig. 7.5: Share of sampled database wait events by type (Stage D).

Dividing the commit rate by throughput gives about 100 database transactions per completed workflow at the knee. A delivery workflow needs on the order of fifteen writing transactions (start, four activity completions, the decisions that follow them, a timer and the final decision); the remainder are polls that found nothing or lost a race for a task. Reducing empty polls is therefore the most direct way to raise capacity on a fixed database.

7.4 Open-Loop Capacity and Overload

Table 7.2 and Fig. 7.6 give the results when workflows were offered at a fixed rate, from half to 1.1 times the closed-loop knee. At 65.6 and 98.3 workflows/s every trial

completed the offered load with a median of 1.5 s and p99 at or below 2.9 s, lower than the floor at $c = 1$. From 108 workflows/s upwards the behaviour became unstable: in most trials completed throughput fell well below the offered rate, thousands of starts were dropped and p99 rose to 40–56 s, although an individual trial at 108 and one at 144 workflows/s kept up.

Table 7.2: Open-loop results (three trials per rate)

Offered (wf/s)	Completed (wf/s)	Dropped	p99 (s)	Lease changes
65.6	64.6–64.9	0	2.2	0
98.3	96.9–97.2	0	2.2–2.9	0
108.2	67.6 / 76.3 / 106.8	3,340 / 2,507 / 0	44.5 / 45.2 / 2.7	10 / 0 / 0
118.0	82–85	2,234–2,648	46–48	10 each
131.1	67–99	2,700–5,889	43–51	10 / 28 / 10
144.2	75.0 / 102.8 / 128.8	5,849 / 3,010 / 0	55.9 / 44.6 / 39.0	14 / 10 / 0

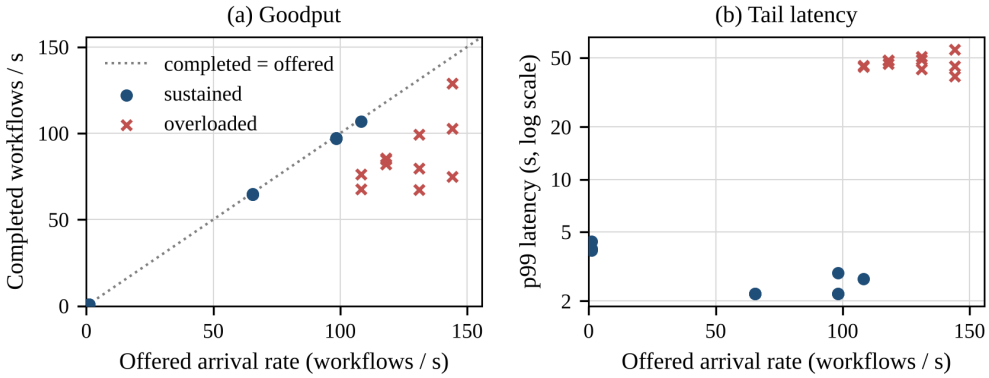


Fig. 7.6: Open-loop offered against completed load (Stage E).

The sustainable open-loop rate is therefore about 100 workflows/s, roughly three quarters of the closed-loop knee. The difference is the one Schroeder *et al.* warn of [35]: a closed loop throttles its own arrivals as latency rises, so it never pushes the system as far past saturation as an open loop does.

The most damaging form of overload was lease loss. In nine of the eleven overloaded trials, one engine lost and re-acquired its five shard leases, and starts sent to those shards in the meantime were refused with `NotOwner`. Closed-loop runs never lost a lease, even at 96% database CPU. Our hypothesis, which these runs do not prove, is that with the pools 60–73% empty and the database above 85% CPU, a renewal waited long enough

for a connection to miss the 10 s lease. If so, lease renewal should be given its own connection, separate from the pool used by workers. In every overloaded trial, safety held: no invariant was violated, refused starts were never accepted, and every accepted run completed. Overload cost availability and latency, not correctness.

7.5 Sensitivity to Workers, Shards and Replicas

Fig. 7.7 and Table 7.3 show the capacity at $c = 256$ as each parameter was varied with the others at their defaults. Worker count has an optimum. With 2+4 workers per shard the database still had headroom (76% CPU) and the workers were the bottleneck; with 8+16 the database spent so much CPU on polling that capacity fell to 120 workflows/s, although more workers did reduce latency at moderate load. At a fixed total of 192 workers, 32 shards performed the same as 16, while 64 shards lost 13%, again because each shard adds polling and renewal work. Spreading the shards more evenly did not help: the ring placed 5, 9 and 2 of the 16 shards on the three nodes, but the limiting resource was the shared database, not the busiest engine.

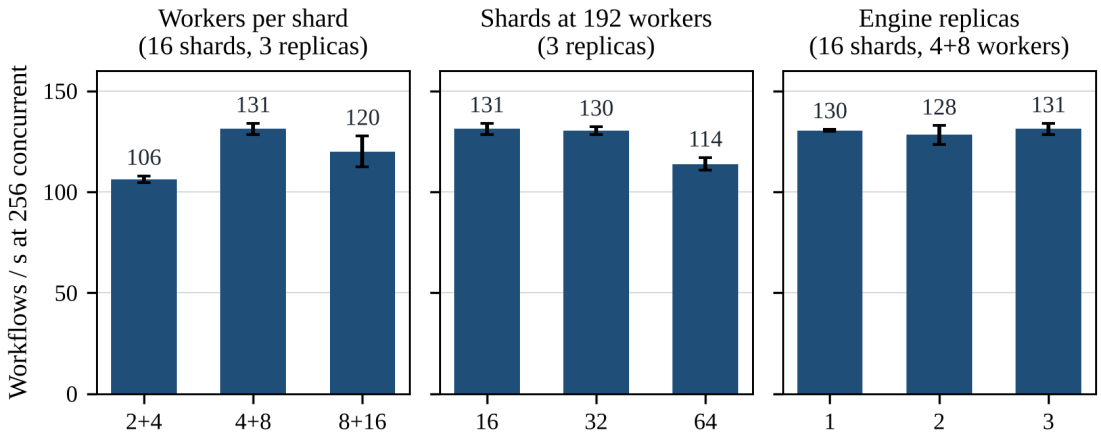


Fig. 7.7: Capacity at $c = 256$ against workers, shards and replicas (Stages F, G and H).

Adding replicas did not increase capacity: one, two and three replicas reached 130.4, 128.2 and 131.1 workflows/s. This is the expected result when all replicas share one database that is already the bottleneck, and it should not be read as a statement about horizontal scalability in general. Past the knee, one replica actually did slightly better (140 against 124 workflows/s at $c = 512$), because its single connection pool limited how many transactions reached the database at once. Measuring horizontal scaling requires a database that is not the limit, which was outside the budget of this project.

Table 7.3: Capacity at $c = 256$ under parameter changes (mean of three trials)

Parameter	Value	Workflows/s	p99 (s)
Workers per shard	2+4	106.2	3.4–4.1
	4+8 (default)	131.1	2.7–3.1
	8+16	120.0	3.8–7.7
Shards (192 workers)	32	130.3	2.8–3.4
	64	113.8	3.9–4.5
Engine replicas	1	130.4	2.7–3.0
	2	128.2	2.6–3.7
	3 (default)	131.1	2.7–3.1

7.6 Recovery from Faults

Table 7.4 and Fig. 7.8 summarise the 16 valid fault trials. The cases fall into three groups.

Table 7.4: Fault recovery under 65.6 wf/s open-loop load

Fault	t_{expire} (s)	t_{claim} (s)	t_{ring} (s)	Refused	p99 (s)
Graceful delete	1.4–4.1	4.1–5.9	31.9–32.3	0–1	4.4–5.8
`SIGKILL`	4.7–5.3	4.7–5.3	34.6–36.8	0–1	2.2–3.3
Freeze 35 s	12.2–12.8	32.3–47.7	51.2–55.2	118–864	31.9–46.2
Database sever	none	none	not broken	0	2.2
Node drain	4.2–8.3	23.1–23.9	38.5–49.7	176–736	4.8–6.7
Database reboot	22.4	33.2	35.2	1,365	5.2

Fast handover. After a graceful delete the victim released its leases, and after `SIGKILL` the restarted pod, which keeps its name in the StatefulSet, re-acquired its own shards. In both cases all shards were held again with a higher token within 4.1–5.9 s, and load errors were at most one start per trial. The remaining 30 s before the cluster was fully healthy were spent by Kubernetes on restarting and readiness, not by the engine.

Slow handover. When an engine was frozen for 35 s, its leases expired after 12.2–12.8 s, the 10 s lease plus up to one renewal interval, but the shards were not claimed by another node until 32–48 s. This is the orphan-adoption delay of Section 4.1.4: a

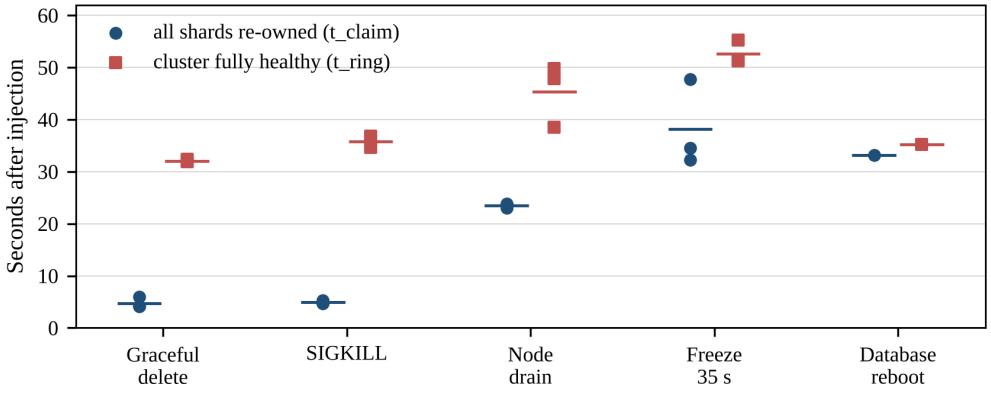


Fig. 7.8: Time until all shards were re-owned and until the cluster was fully healthy, per fault.

survivor adopts a shard only after its lease has been expired for more than twice the lease period. A drained node's shards were re-owned in about 23 s, and the database reboot took 33 s for all 16 shards. In these cases, starts sent to an unowned shard were refused (118 to 1,365 per trial), and for the freeze p99 rose to 32–46 s. The adoption delay was chosen to prevent needless handovers after short pauses; these results show its cost, and a shorter delay, or one that depends on whether the owner answers its health check, would reduce unavailability after a real failure.

No handover. Severing an engine's database connections did not move any lease, because the engine reconnected within its renewal interval; progress on its shards resumed within a few milliseconds.

The most important result concerns safety. When a frozen engine resumed, it still believed it owned its shards, and it attempted to write to them. Those writes were rejected by the fencing check 22, 4 and 5 times in the three trials. Across all 16 valid trials no journal invariant was violated: there were no duplicate sequence numbers, duplicate activity completions, sequence gaps or runs with more than one terminal event, and every accepted run completed after the load drained. Fencing therefore did what it was designed to do on real infrastructure.

One engine defect was found. During one drain trial, the restarted engine's start-up migration deadlocked with the other engines' live traffic; the process exited and Kubernetes restarted it, after which the trial recovered normally. Migrations should take an advisory lock and be skipped when the schema is already current.

7.7 Correctness Summary

Over the closed- and open-loop stages, about 140,000 workflow runs were checked by the journal invariants after their measurement points, together with every run in the fault trials and the local duplicate audit. No violation was found in any of them. This is evidence for these workloads and faults; it is not a proof, and the runs did not include network partitions between engines, which were not tested.

7.8 Semantic Deduplication

Table 7.5 gives the results of the offline evaluation of Section 6.3. Exact matching never suppresses a distinct request, but it detects only the 14 of 98 duplicates that are byte-identical after normalisation. Pure semantic matching at the shipped thresholds ($\tau_{\text{low}} = 0.85$, $\tau_{\text{high}} = 0.90$) detected 56 duplicates automatically and sent 25 pairs to a person, but it also judged 108 of the 115 distinct pairs to be duplicates, including all 91 lexically close ones. No single threshold between 0.70 and 0.99 separated the classes; the encoder places "BLR to DEL Friday morning" and "BLR to DEL Saturday morning" as close together as genuine paraphrases. The reworded double-booking case of Section 1.2.3 had a similarity of 0.87, inside the escalation band.

Table 7.5: Deduplication on the 213-pair corpus at the shipped thresholds

Comparator	Detected	False positives	Escalated	Precision	p50 / p99 (μs)
Exact	14 / 98	0 / 115	0	1.000	2.4 / 8.2
Semantic	56 / 98	108 / 115	25	0.341	10.1 / 66.2
Hybrid (opt-in)	98 / 98	0 / 115	0	1.000	14.8 / 81.4

The hybrid comparator, which compares the deciding fields exactly after normalisation and uses similarity only for the rest, detected all 98 duplicates with no false positive at every threshold tested. This result must be read with two cautions. The corpus was built by construction and its labels were not assigned by people, and the same construction defines which fields are deciding; a perfect score on such a corpus shows that the mechanism works as intended, not that it would score as well on real agent traffic. The thresholds were also not chosen on a held-out set. The comparison cost, excluding the embedding call, was below 100 μs at p99 for every comparator, negligible beside a decision commit.

The conclusion we draw is that a general-purpose sentence encoder on its own is not a safe basis for suppressing actions, because it measures topical similarity, and a change of one word can change the identity of an action. The design decisions of Section 4.2.6, journalled verdicts, an escalation band and opt-in use, limit the damage of a wrong verdict, but the hybrid comparator is what makes the mechanism usable.

7.9 Threats to Validity

Several factors limit how far these results can be generalised. All measurements were taken in one availability zone, on one day per campaign, on burstable "flex" instances whose performance can vary over time; trials were repeated three times but not across days. Each measurement point lasted two minutes after warm-up, which is long enough to observe saturation but not long-term effects such as table bloat or autovacuum. The database was a single 2-vCPU host, so every capacity figure is a property of that host, and nothing here bears on database high availability. Only one workflow type was used for load. Finally, the semantic corpus was synthetic, as discussed above.

CHAPTER - 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project set out to build a durable execution engine that partitions long-running workflows across several nodes, survives the failure of any of them without losing progress or repeating side effects, and needs no infrastructure other than PostgreSQL. The engine records every decision of a workflow in an append-only history and recovers a run by replaying its code against that history. Workflows are placed on a fixed set of shards, and shards on nodes by a consistent-hash ring, so that ownership can move without moving data. Ownership is held through a lease row per shard, and every transaction that writes shard state re-checks the shard's fencing token under a row lock before it commits.

The evaluation on Amazon EKS supports the main claims within the limits set out in Section 7.9. Under sixteen valid fault trials of six kinds, including a 35 s freeze of an engine process that outlived its lease, no journal invariant was violated and every accepted run completed; the frozen engine's attempts to write after it resumed were rejected by fencing in every trial. After a graceful deletion or a process kill, all of a node's shards were re-owned within 6 s. The cost of faults was availability, in the form of refused starts and longer tail latency while shards were unowned, and never correctness.

The evaluation also located the limits of the design. On a single 2-vCPU database host the engine sustained about 131 workflows per second in a closed loop and about 100 in an open loop, and adding engine replicas did not change this, because the database was the bottleneck in every configuration. Much of the database's work was polling: an idle cluster committed about 250 transactions per second, and at the knee about 100 transactions were spent per completed workflow. Polling also set a latency floor of about 2.2 s for a workflow with four activities and a short timer. Under open-loop overload the system lost shard leases, which turned overload into unavailability. And a general-purpose sentence encoder proved to be a poor basis for semantic deduplication on its

own, judging most distinct but similar requests to be duplicates; comparing the deciding fields of a request exactly, and using similarity only for the remainder, resolved this on the evaluation corpus.

8.2 Future Work

The results point to the following work, in roughly the order of its expected benefit.

1. *Notification instead of polling.* Waking workers with PostgreSQL `LISTEN/NOTIFY` when tasks and timers become ready, and polling only as a fallback, should lower both the idle load and the latency floor, and raise capacity on the same database.
2. *Isolated lease renewal.* Renewing leases on a dedicated connection, outside the pool used by workers, should prevent the lease loss seen under open-loop overload. Admission control that refuses work before the pool is exhausted would complement it.
3. *Faster adoption of orphaned shards.* Making the adoption delay depend on whether the owner answers its health check would shorten unavailability after a freeze or a drain without causing churn after short pauses.
4. *Safe migrations.* Migrations run under an advisory lock, and skipped when the schema is current, would remove the start-up deadlock.
5. *Scaling and availability of the database.* Repeating the evaluation with a larger database host, and with a synchronous standby or a managed highly available database, is needed to measure horizontal scaling of the engine itself and to test failover of the database.
6. *History retention.* Histories of completed runs should be archived or compacted.
7. *Semantic deduplication on real traffic.* The hybrid comparator should be evaluated on agent traces labelled by people, with thresholds chosen on a held-out split, and the escalation path should be studied as a human-in-the-loop interaction.
8. *Wider fault coverage.* Network partitions, packet loss, instance termination and rolling upgrades under load remain untested.

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